A moment worth practicing

CadenceLab is a Track C speech evidence workbench for student speakers and coaches. It compares two recordings of the same words and turns acoustic differences into timed, explainable feedback. Users can play either interval, inspect numerical evidence and export JSON. Scores describe similarity to the selected reference, not talent, truth or rhetorical quality.

Source: README, docs/DATASET.md, docs/SECURITY.md and evaluation.json

A labeled contrastive corpus

The corpus contains 90 paired cases from five excerpts, two speakers and three public archival speeches. Five perturbations each have three severity levels: pace, energy, inserted pauses, Gaussian noise and local pitch. Three controls test unchanged audio, global gain and global pitch. Injection intervals derive from exact transformation sample boundaries independently of detector predictions. Rights, hashes and construction code are documented in DATASET.md.

Source: README, docs/DATASET.md, docs/SECURITY.md and evaluation.json

The technical path

A pinned 95 MB quantized Wav2Vec2 model produces token probabilities. Transcript-constrained CTC Viterbi alignment maps characters and words to acoustic frames; timestamps are not evenly allocated. Four-second windows have 0.5-second context. Original audio is preserved for evidence; automatic gain control affects alignment only. Browser analysis uses an ONNX WebAssembly worker with no audio-upload endpoint.

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Evidence and uncertainty

Energy is normalized per recording; pitch is centered in semitones and offset-corrected across matched words. Matched-word windows compare pace. FFT, autocorrelation and spectral flatness characterize texture and pitch; Python additionally computes 13 MFCCs. Poor phonetic support with measurable noise yields a partial texture report with no word-based score. Unsupported alignment abstains. Temporal intervals retain the recordings own time axes.

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Evaluation, including the misses

43 of 75 altered cases localized the target change at union temporal IoU >= 0.30. Moderate and strong cases: 40 of 50. Mean temporal IoU including misses: 0.403. Two of fifteen controls produced false regions; ten unchanged/global-gain controls produced none. Four cases abstained completely and two produced partial reports. These are development-set diagnostics, not held-out estimates. Historical English male speakers do not establish real-student or demographic validity.

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Verification and reproducibility

Five pipeline tests pass, including repeated-character CTC behavior, normalization and all 90 audio hashes and intervals. Automated checks found no axe accessibility violations across four screens, no mobile overflow and no page errors. A genuine browser upload recovered three energy intervals; Python/browser score agreement was 94.8 with <=20 ms boundary differences. README covers setup, model pin, rights, limitations and AI-tool disclosure. Future work requires consent-based real-student validation, broader accents/devices and an independent test set.

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